Formulas

discrete time growth:

- $N_T = N_0 \lambda^T$
- $\lambda = f + p$
- $\mathcal{R} = f/(1-p)$

continuous time growth:

- $N(t) = N(0)\exp(rt)$
- $r = b - d$
- $\mathcal{R} = b/d$

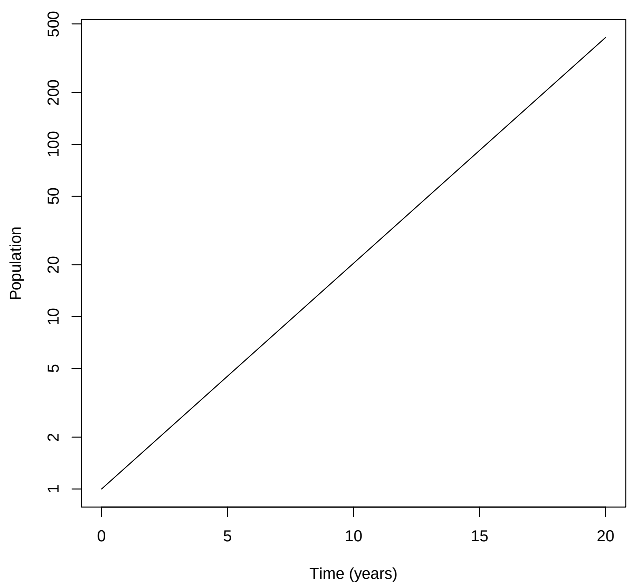
structured growth:

- $\ell_x = p_1 \times p_2 \times \ldots p_{x-1}$
- $\mathcal{R} = \sum \ell_x f_x$

competition:

- α_{ij} = effect **of** species i **on** species j
- $C = \alpha_{12}\alpha_{21}$
- $E_{ij} = \alpha_{ij} K_i / K_j$

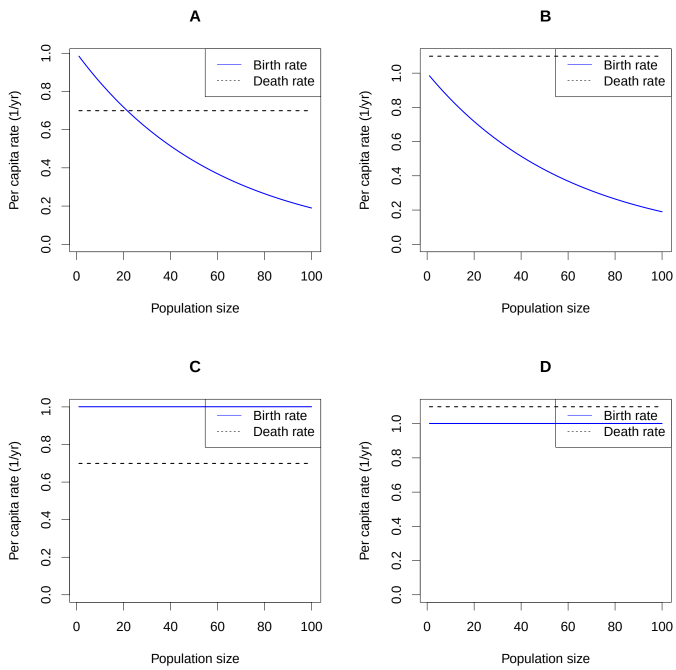
Use the picture below for the next two questions. It shows a time series for a continuous-time birth-death model.



1. This picture shows a population that is:

A. Increasing arithmetically
B. Increasing geometrically
C. Increasing arithmetically on the log scale, but geometrically on a linear scale
D. Increasing geometrically on the log scale, but arithmetically on a linear scale

2. Which of the small pictures shows the assumptions that generated the time plot?



A. A
B. B
C. C
D. D

3. A species of small plant is highly adapted to dramatic changes in conditions (ranging from very good growth conditions to very bad). We would expect it to:

A. Have a very fast life cycle
B. Have a very slow life cycle
C. Have adaptations that speed up its life cycle in good conditions and slow it down in bad conditions
D. Have adaptations that speed up its life cycle in bad conditions and slow it down in good conditions

4. A simple population model has *structure*, but not *regulation* (individuals are assumed to be independent). Cohorts are not independent. If the model has $\mathcal{R}_0 > 1$, then: The modeled population __________ grow exponentially at first, and __________ grow exponentially as it approaches a stable age distribution (SAD)

A. will; will
B. may not; will
C. will; may not
D. may not; may not

5. We believe that in a particular desert, Joshua trees and cactuses are competing only for water. According to our simple theory, we would expect:

A. A tendency for co-existence
B. The species with the larger value of K (indiv/ha) to dominate
C. The species with the larger value of α (measuring its impact on the other species) to dominate
D. The species with the larger value of E (measuring its impact on the other species) to dominate
E. A tendency for mutual exclusion

6. A scientist has made a life table for a perennial population of deer, with censusing *before* reproduction. If she re-does the life table with a census after reproduction, she would expect $\mathcal{R}$ to be __________ because population dynamics look __________ when viewed before reproduction.

A. higher; fastest
B. higher; slowest
C. lower; fastest
D. lower; slowest
E. the same; the same

7. Which of these is *not* an effect that tends to *balance* the tendency of populations to grow or decline exponentially without limit?

A. Allee effects
B. Competition for access to mates
C. Competition for access to resources
D. Resource depletion
E. Dynamics of natural enemies

8. Which of the following is *not* a possible advantage of producing fewer offspring with the same total amount of energy (thus investing more in each offspring)?

A. Greater ability to help offspring survive
B. Greater ability to help offspring disperse
C. Greater ability to produce high-quality offspring
D. More energy left over to produce offspring in future years

9. Japanese maples are trying to invade a stable Ontario forest dominated by sugar maples (we don't know whether they will be successful). Based on these assumptions, we would expect the reproductive number of the ________ population to satisfy ________.

A. sugar maple; $\mathcal{R} > 1$
B. Japanese maple; $\mathcal{R} > 1$
C. sugar maple; $\mathcal{R} = 1$
D. Japanese maple; $\mathcal{R} = 1$

Use the following information for the next two questions. A population of hemlock trees is estimated to be at stable age distribution, with a constant life table, with reproductive number $\mathcal{R}$= 1.2. It takes the trees several decades to reach maturity and reproduce.

10. This population is

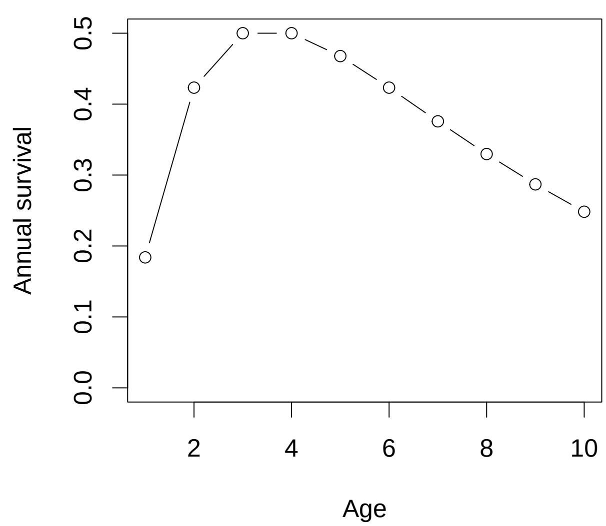
A. declining
B. stable
C. increasing
D. showing damped oscillations
E. there is not enough information to answer this question

11. What is the *most accurate* statement you can make about the finite rate of growth λ, measured with a time step of one year?

A. We expect $\lambda > 1.2$
B. We expect $\lambda = 1.2$
C. We expect $\lambda > 1$
D. We expect $1.2 > \lambda > 1$
E. We expect $1.2 > \lambda$

12. The "balance argument" for sexual allocation implies that, if individuals do not interact strongly with their siblings, females should on average:

A. Produce the same number of male and female offspring
B. Use the same amount of resources *per offspring* for male and female offspring
C. Use the same *total* amount of resources for male and female offspring
D. All of the above

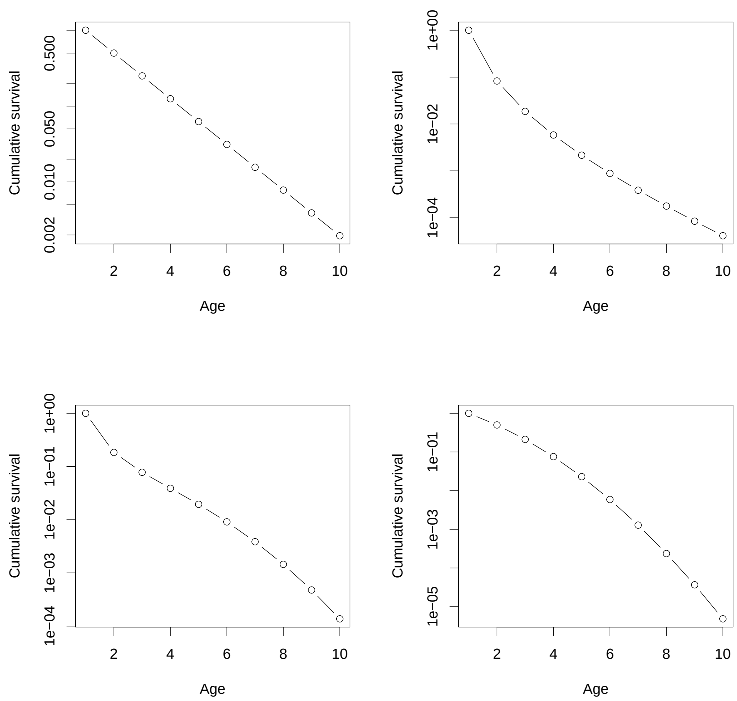


Use the picture above for the following 2 questions.

13. What does this picture of survivorship in an idealized age-structured population indicate about *mortality* in this population?

A. Mortality is constant
B. Mortality is elevated in older individuals
C. Mortality is elevated in younger individuals
D. Mortality is elevated in both older and younger individuals

14. The pictures below show *cumulative* survival. Which one corresponds to the picture shown above?



A. Upper left
B. Upper right
C. Lower left
D. Lower right

15. Which of the following traits is *not* typically associated with r strategies?

A. Fast life cycle
B. Efficient resource use
C. Relatively small investment per offspring
D. Relatively large investment in dispersal

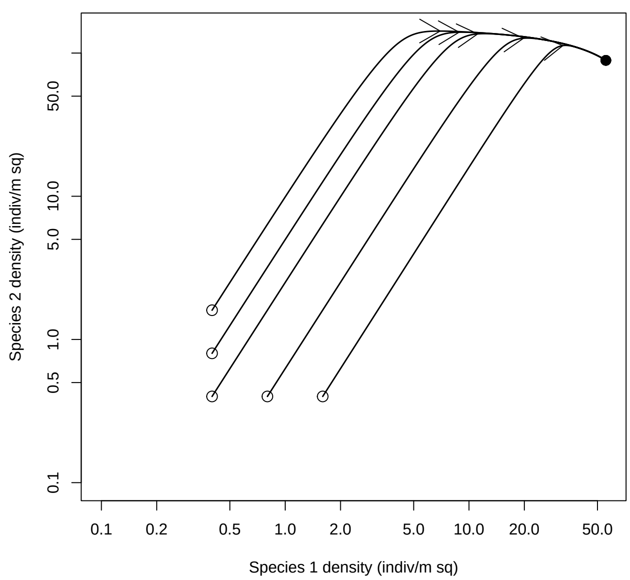
16. In an environment where seed survival is highly variable from year to year, we expect that semelparous plants (reproducing only once) are likely to invest relatively __________ than iteroparous plants in __________.

A. more; mechanisms for seed dispersal
B. less; mechanisms for seed dispersal
C. more; each individual seed
D. less; each individual seed

17. We expect populations to be regulated (kept under control). This typically works because their __________ tends to decrease when the population __________.

A. growth rate; has been established for a long time
B. growth rate; becomes larger
C. projected future size; has been established for a long time
D. projected future size; becomes larger

Use the picture above for the next three questions. The path in the middle corresponds to a starting density of 0.4indiv/m^2 for each species.



18. This picture shows what sort of competition outcome?

A. Species 1 dominates
B. Species 2 dominates
C. The species co-exist
D. There is mutual exclusion

19. Assuming density dependence can be neglected at low densities, which species has the higher value of r_{max}?

A. Species 1
B. Species 2
C. They have the same value of r_{max}
D. Either species can have a higher value, depending on initial conditions.

20. From this picture, we can conclude that:

A. Both of the competition coefficients $\alpha < 1$.
B. Both of the competition coefficients $\alpha > 1$.
C. Both of the population-level competitive effects $E < 1$.
D. Both of the population-level competitive effects $E > 1$.
E. We cannot make any of these conclusions from this picture.

Short-answer questions

Answer questions *in pen. Briefly* show necessary work and equations. Points may be *deducted* for wrong information, even when the correct information is also there.

21. An annual plant has variable fecundity: each female has an average of 3 successful female offspring in an good year (half of the time) and an average of 0.4 successful female offspring in a bad year (the other half of the time).

a) (1 point) What is the long-run average λ for this species?

b) (1 point) What are two possible strategies this plant could evolve to spread out the risk of a bad year (hedge its bets)?

22. A species of falcon and a species of hawk are living in the same region and competing to eat songbirds and small mammals that they find there.

a) What is a possible reason why the competition between species might be relatively *stronger* than within species in this case?

b) What is a possible reason why the competition between species might be relatively *weaker* than within species in this case?

c) What are *two* possible outcomes of the competitive interaction in the second case (when between-species competition is weak)?